

Olist E-Commerce Business Performance Analysis

A Data-Driven Case Study | 2016-2018

Jacinth Samuel CK | Power BI | Kaggle Public Dataset

Executive Summary

Between 2016 and 2018, Olist — Brazil's largest e-commerce marketplace — processed approximately 99,000 orders generating around \$16M in Gross Merchandise Value across 27 Brazilian states. The business demonstrated exceptional top-line growth, with monthly revenue scaling from approximately \$138K in January 2017 to over \$1.1M in January 2018 — a year-over-year increase of more than 700% in a single month.

However, beneath the growth story lies a structural problem that threatens long-term sustainability: a 3.12% repeat purchase rate. In an industry where the benchmark sits at 25-30%, Olist was acquiring customers at scale but failing to retain them. 97 out of every 100 customers never came back.

This case study analyzes Olist's performance across five dimensions — sales growth, customer behavior, geographic intelligence, operational efficiency, and product economics — and concludes with strategic recommendations including projected financial outcomes.

1. Business Context

What Olist Is

Olist is not a retailer. It is a marketplace — a platform that connects small and medium-sized Brazilian businesses to customers across the country. Sellers list products on Olist's platform, customers place orders, and third-party logistics partners handle delivery. Olist earns revenue through seller commissions (approximately 12-15% per transaction) and monthly subscription fees.

This distinction matters analytically. The ~\$16M figure in this analysis represents platform GMV — the total value of all transactions processed through Olist. Olist's actual earned revenue is estimated at approximately \$1.9-2.4M over this period. Any analysis that treats GMV as Olist's profit would be fundamentally misleading.

The Dataset

Dataset	Records	Key Information
Orders	~99,000	Order dates, status, delivery timelines

Dataset	Records	Key Information
Order Items	~112,000	Products, prices, freight costs
Customers	~99,000	Location, unique customer identity
Payments	~103,000	Payment method, installments, values
Reviews	~99,000	Customer satisfaction scores
Products	~33,000	Categories, dimensions, weight

Seven tables were joined using a star schema data model in Microsoft Power BI, with the orders table serving as the central fact table.

2. Sales Performance Analysis

Revenue Growth

Period	Monthly Revenue	Notes
January 2017	~\$138K	Early platform stage
November 2017	~\$1.19M	Peak month — Black Friday effect
December 2017	~\$878K	Post-holiday dip
January 2018	~\$1.1M	700%+ YoY growth vs Jan 2017

The November 2017 peak of approximately \$1.19M represents the strongest single month in the dataset — driven by Brazil's Black Friday shopping event. December's dip to around \$878K is a natural post-holiday correction seen across all e-commerce markets globally.

What is most significant is January 2018 at over \$1.1M — nearly matching the November 2017 peak without any seasonal event driving it. By early 2018, Olist had structurally elevated its baseline monthly revenue rather than just achieving seasonal spikes.

Order Volume Growth

Period	Monthly Orders	Notes
January 2017	~800 orders	Early platform stage
November 2017	~7,500 orders	Peak month
December 2017	~5,700 orders	Post-holiday dip

Period	Monthly Orders	Notes
January 2018	~7,300 orders	Strong baseline established

The parallel growth in both revenue and order volume at a consistent average order value of \$161 indicates growth was driven by new customer acquisition rather than increasing spend per customer — important context for the retention analysis ahead.

Top Revenue Categories

Rank	Category	Approx Revenue
1	Health & Beauty	~\$1.25M
2	Watches & Gifts	~\$1.2M
3	Bed, Bath & Table	~\$1.04M
4	Sports & Leisure	~\$950K
5	Computer Accessories	~\$880K
10	Tools & Garden	~\$485K

Health & Beauty leads at approximately \$1.25M — a strong but not dominant lead over second-place Watches & Gifts at around \$1.2M. The gap between first and tenth place represents a 2.6x revenue difference — meaningful category concentration but not extreme.

Order Fulfillment

Of approximately 99,000 orders placed, around 96,500 (96.48%) were successfully delivered — a strong fulfillment rate for a marketplace at this scale. The roughly 630 canceled orders at \$161 average order value represent approximately \$100K in lost GMV — translating to around \$12-15K in lost Olist commission revenue.

3. Geographic Analysis

Revenue by State

Rank	State	Approx Revenue	% of Total
1	São Paulo (SP)	~\$6M	~37%
2	Rio de Janeiro (RJ)	~\$2.1M	~13%

Rank	State	Approx Revenue	% of Total
3-10	Other states	~\$8M	~50%

The drop from São Paulo (~\$6M) to Rio de Janeiro (~\$2.1M) is stark — SP generates nearly 3x more revenue than the second-highest state.

Revenue by City

Rank	City	Approx Revenue
1	São Paulo	~\$2.2M
2	Rio de Janeiro	~\$1.16M

São Paulo city alone drives approximately \$2.2M — nearly 14% of total platform GMV. The top 2 cities combined account for over \$3.3M — more than 20% of all revenue across the entire platform.

Geographic concentration risk: Olist's dependence on São Paulo for such a large share of revenue creates real vulnerability. Any economic disruption, competitive entry, or regulatory change specifically targeting São Paulo could have outsized platform-wide impact. Diversification into MG, RS, and PR — which show healthy secondary activity — is a strategic priority.

4. Customer & Payment Analysis

Payment Behavior

Around 78% of all transactions were completed by credit card. Boleto (Brazilian bank payment slip) accounted for approximately 18%. The average number of payment installments was approximately 2.6 throughout the analysis period — stable across all three years with no significant upward trend.

Cash flow implication: At 2.6 average installments, most customers split payments across approximately 3 months. Olist ships the product in month 1 but collects full payment by month 3. This creates a consistent 2-3 month cash flow lag — manageable today but increasingly significant as the platform scales.

The Retention Crisis

Repeat purchase rate: 3.12%

This is the single most important finding in the entire analysis.

Of approximately 96,000 unique customers acquired across 2 years:

- Around 93,000 customers (96.88%) placed exactly one order and never returned
- Around 3,000 customers (3.12%) made at least one repeat purchase

Monthly repeat orders averaged approximately 300 per month throughout the entire dataset period. While new customer orders grew significantly — from roughly 800 in January 2017 to over 7,000 in January 2018 — repeat orders remained flat at around 300 per month. The two lines never converge. Retention was not improving despite rapid platform growth.

Industry context:

- Top performers (Amazon, Flipkart): 40–50% repeat rate
- Average marketplace: 25–30% repeat rate
- Olist: 3.12%

The gap between Olist and average industry performance represents an enormous untapped revenue opportunity hiding in plain sight.

5. Operations Analysis

Delivery Performance by State

State	Avg Delivery vs Estimate	Performance
PR (Paraná)	~42 days early	Best performing state
SP (São Paulo)	~41 days early	Strong performance
BA (Bahia)	~32–33 days early	Fast but low satisfaction
ES (Espírito Santo)	~21 days late	Only late-delivering state

Paraná is the strongest performing delivery state — orders arrive approximately 42 days ahead of estimate. São Paulo follows closely at around 41 days early. Espírito Santo is the sole outlier — consistently late by approximately 21 days while every other state delivers ahead of schedule.

This pattern almost certainly indicates a specific logistics partner issue in ES rather than a systemic Olist problem.

The Delivery-Satisfaction Paradox

State	Delivery Performance	Avg Review Score
BA	~33 days early	3.86 (lowest)

State	Delivery Performance	Avg Review Score
ES	~21 days late	4.04
SP	~41 days early	4.17
PR	~42 days early	4.18 (highest)

The most counterintuitive finding in this analysis: Bahia receives orders approximately 33 days ahead of schedule — yet records the lowest customer satisfaction score at 3.86. Espírito Santo delivers 21 days late — yet scores a reasonable 4.04.

This definitively disproves the hypothesis that delivery speed drives customer satisfaction on Olist. Product quality, accurate product descriptions, and seller communication appear to be stronger drivers of the review score than delivery timing.

Conclusion: Olist's improvement strategy should prioritize seller quality standards over logistics optimization. Investing only in faster delivery without addressing product quality would have limited impact on satisfaction scores.

6. Product Economics Analysis

Freight Cost as Percentage of Product Price

Category	Freight to Price Ratio	Implication
Casa Conforto (Home Comfort)	~53%	Critical — conversion killer
Flores (Flowers)	~44%	High — perishable handling
Moveis Colchao (Mattresses)	~38%	Expected for large items
PCs (Computers)	~4.4%	Excellent — high value product

A ~53% freight-to-price ratio for home comfort products means customers pay approximately 53 cents in shipping for every dollar of product. When a customer sees a \$20 product with a \$10+ shipping charge, cart abandonment becomes the most likely outcome.

PCs represent the healthiest freight economics at approximately 4.4% — high product value makes shipping costs negligible relative to price.

Product Weight vs Freight Correlation

Analysis confirmed a positive correlation between product weight and freight cost. Categories below 2,000g average weight cluster around \$15–20 in average freight. Categories exceeding

8,000g show freight costs of \$35-45 — roughly double the platform average. This correlation is logical and the pricing appears broadly proportional across most categories.

7. Strategic Recommendations & Projected Outcomes

Recommendation 1 — Customer Retention Program

Priority: Critical

The problem: ~3% repeat rate vs 25-30% industry benchmark.

Proposed actions:

- Post-purchase email sequence: delivery confirmation → 7-day review request → 30-day re-engagement discount
- Basic loyalty points program: earn points per purchase, redeem for discount
- Personalized campaigns targeting top 10% of customers by spend

Projected outcome: Moving repeat rate from ~3% to 8% would mean approximately 7,700 additional repeat customers. At ~\$161 average order value, this generates roughly \$1.24M in additional GMV — approximately \$185K in additional Olist commission revenue annually. A more aggressive 15% repeat rate target would generate over \$2M in incremental GMV.

Timeline: 6-12 months to see meaningful improvement.

Recommendation 2 — Geographic Diversification

Priority: High

The problem: São Paulo drives ~37% of revenue. Single-state concentration is a business risk.

Proposed actions:

- Recruit sellers specifically serving MG, RS, and PR markets
- Reduced commission rates for first 6 months for sellers in underrepresented states
- Targeted marketing in Belo Horizonte, Porto Alegre, and Curitiba

Projected outcome: If MG, RS, and PR each grew to match RJ's ~\$2.1M revenue level, this would add approximately \$3.5M in incremental GMV — a ~22% increase in total platform revenue. Reducing SP concentration from 37% to 25% would also significantly reduce geographic risk.

Timeline: 12-18 months for meaningful rebalancing.

Recommendation 3 — Freight Cost Optimization

Priority: Medium

The problem: ~53% freight-to-price ratio in home comfort category causing cart abandonment.

Proposed actions:

- Negotiate category-specific logistics rates for high-ratio categories
- Freight subsidy for products priced below \$15 where freight ratio exceeds 40%
- Free shipping threshold at \$50+ to encourage higher basket sizes

Projected outcome: A conservative 15% reduction in cart abandonment for high-ratio categories could generate approximately \$360K in incremental GMV annually — around \$54K in additional Olist commission.

Timeline: 3-6 months for renegotiation and threshold implementation.

Recommendation 4 — Seller Quality Program

Priority: Medium

The problem: BA state delivers fastest but scores lowest. Product quality drives satisfaction more than speed.

Proposed actions:

- Seller quality scoring based on review scores not just fulfillment speed
- Mandatory product listing quality training for sellers averaging below 4.0
- Surface seller review scores prominently during product discovery

Projected outcome: A 0.2 improvement in average review score (from 4.09 to ~4.3) correlates with approximately 3-5% improvement in conversion rate industry-wide. Applied to ~99,000 orders this could generate approximately \$800K in incremental GMV annually.

Timeline: 6-9 months to see review score improvement at scale.

Recommendation 5 — Espírito Santo Logistics Audit

Priority: Medium

The problem: ES is the only state with consistent late delivery — ~21 days behind estimate.

Proposed action: Audit the specific logistics partner serving ES. If SLA performance is below contract terms, escalate or replace.

Projected outcome: Bringing ES in line with platform average would improve customer satisfaction in the state and eliminate the only consistent late-delivery geography on the platform. Low cost, high impact.

Timeline: 1–3 months.

8. Combined Impact Summary

Recommendation	Incremental Annual GMV	Incremental Olist Revenue
Retention Program	~\$1.24M	~\$185K
Geographic Diversification	~\$3.5M	~\$525K
Freight Optimization	~\$360K	~\$54K
Seller Quality Program	~\$800K	~\$120K
ES Logistics Audit	Modest but immediate	Modest but immediate
Total	~\$5.9M	~\$884K

If all recommendations are implemented successfully, projected incremental annual GMV could reach approximately \$5.9M — a 37% increase over baseline. Olist's annual commission revenue could increase by approximately \$885K.

These are conservative estimates based on industry benchmarks and the specific data patterns identified in this analysis.

9. Data Limitations & Methodology

- Dataset covers September 2016 to August 2018 — post-August figures in monthly charts reflect 2017 data only
 - \$16M = GMV not Olist's actual earned revenue
 - Repeat rate calculated using customer_unique_id not customer_id
 - Freight analysis used price from order_items (product level) not payment_value (order level)
 - Tool: Microsoft Power BI Desktop with star schema data model and 6 custom DAX measures
-

10. Conclusion

Olist demonstrated exceptional growth between 2016 and 2018. Monthly revenue grew over 700% year-on-year. Order volume scaled nearly 10x. Around 96.5% of orders were successfully delivered with a 4.09/5 satisfaction score. The operational foundation is genuinely solid.

But the business is running a leaking bucket. It acquires customers efficiently and delivers products reliably — then loses 97% of those customers permanently. The five recommendations in this analysis — with a combined projected GMV impact of approximately \$5.9M annually — represent a clear, data-backed path from growth to sustainable profitability.

The retention program is the highest priority. Everything else builds on a returning customer base.

Jacinth Samuel CK / Business Performance Case Study / Microsoft Power BI / Kaggle Public Dataset / 2026